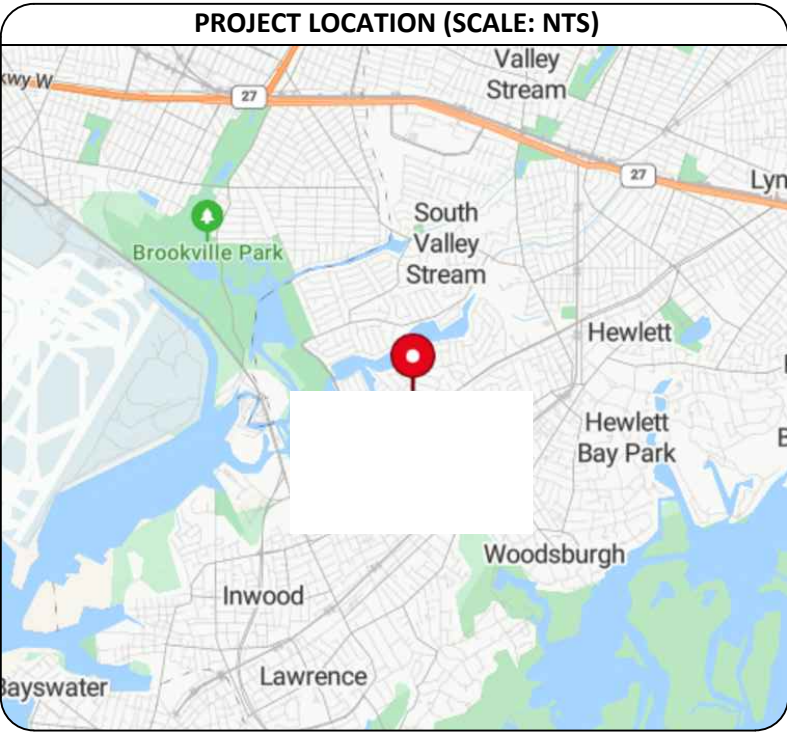
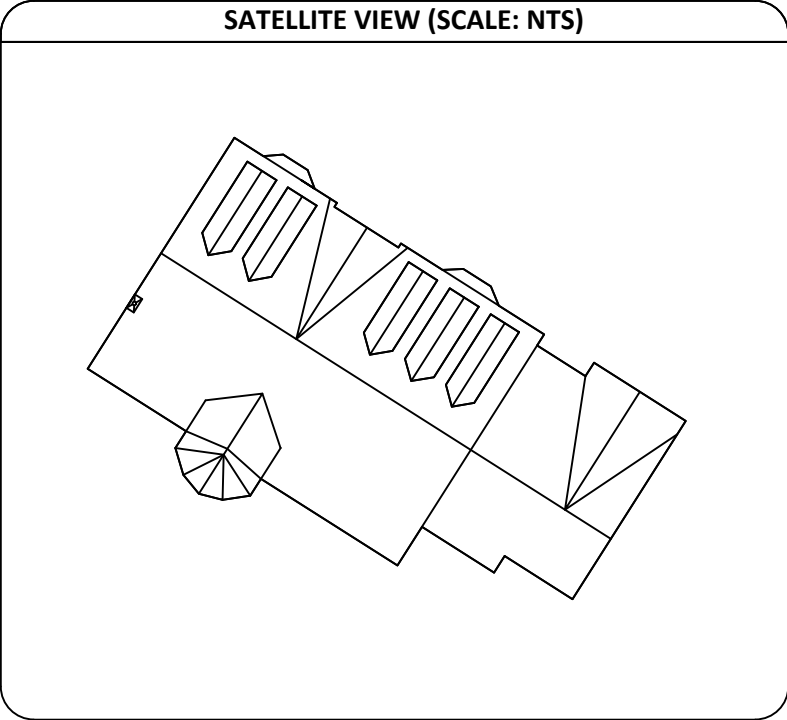


PROJECT: PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(31) MODULES x 425W	(31) MICROINVERTERS x 325W
13.18 kW	10.08 kW



- SCOPE OF WORK
- PROPOSED PROJECT TO INSTALL GRID- TIED PV SYSTEM ON AN EXISTING ROOFTOP STRUCTURE AT A RESIDENTIAL LOCATION: 1234 STREET NAME, CITY, NY 9874. COORDINATES: 40.638282, -73.72221
 - THE PV MODULES WILL BE SECURED ON ROOF USING PRE-ENGINEERED ROOF RACKING SYSTEM.
 - INSTALLED PV SYSTEM WILL BE CONNECTED TO HOMEOWNER'S SERVICE EQUIPMENT USING APPLICABLE INTERCONNECTION METHOD (AS PER AHJ).
 - THE ENERGY PRODUCED BY PV SYSTEM WILL BE USED TO POWER THE HOME APPLIANCES. EXCESS POWER IS SENT BACK TO THE UTILITY GRID.

- APPLICABLE GOVERNING CODES & NOTES
- THIS PROJECT SHALL COMPLY WITH THE FOLLOWING CODES:
- 2020 NEW YORK BUILDING CODE
 - 2020 NEW YORK RESIDENTIAL CODE
 - 2020 NEW YORK MECHANICAL CODE
 - 2020 NEW YORK PLUMBING CODE
 - 2020 NEW YORK ENERGY CODE
 - 2020 NEW YORK FIRE CODE
 - 2017 NEW YORK ELECTRICAL CODE
- ALL OTHER RELATED STATE & AHJ BY-LAWS & ORDINANCES

- DESIGN CRITERIA
- CONSTRUCTION: SINGLE FAMILY DWELLING (SFD)
 - NO. OF STORY: DOUBLE STORY
 - EXPOSURE CATEGORY : B, RISK CATEGORY : II
 - WIND SPEED (ASCE 7-16) : 125 MPH
 - GROUND SNOW LOAD (ASCE 7-16) : 25 lb/sqft
 - ASHRAE EXTREME LOW: -14°C
 - ASHRAE HIGH TEMP (2% AVG.) : 31°C
 - ASHRAE DISTANCE ABOVE ROOF (7/8") : 53°C

SHEET INDEX	
PV-0 & PV-0.1	PROJECT SUMMARY
PV-1	SITE PLAN
PV-2	ATTACHMENT DETAILS
PV-3	BOM & STRING LAYOUT
S-1	STRUCTURAL DETAILED DIAGRAM
S-2	EQUIPMENT ELEVATION DIAGRAM
E-1	ELECTRICAL DIAGRAM & CALCULATIONS
E-2	PLACARDS
E.S	EQUIPMENT SPECIFICATIONS

EQUIPMENT DETAILS	
(#) PV MODULES	(31) TRINA SOLAR TSM-NE09RC.05 425
(#) MICRO- INVERTER	(31) ENPHASE IQ8M-72-2-US
(#) COMBINER BOX	(01) ENPHASE IQ COMBINER 5/5C (X-IQ-AM1-240-5/5C)
OUTPUT CURRENT	41.85A
BRANCH CIRCUIT	2 BRANCH CIRCUITS OF 10 MODULES & 1 BRANCH CIRCUIT OF 11 MODULES
ROOF RACKING SYSTEM	IRONRIDGE HALO ULTRA GRIP / IRONRIDGE AIRE A1 FLUSH MOUNT
POINT OF INTERCONNECTION	LINE SIDE TAP IN THE INTERIOR MAIN SERVICE PANEL WITH 200A MAIN DISCONNECT, 200A BUSBAR, 120/240V, 3W, 1φ
UTILITY	PSEG
AHJ	TOWN OF HEMPSTEAD

- GENERAL NOTES
- ALL COMPONENTS ARE UL LISTED & NEC CERTIFIED, WHERE WARRANTED
 - A DISCONNECTING MEANS SHALL BE INSTALLED AT A READILY ACCESSIBLE LOCATION EITHER OUTSIDE OF THE BUILDING OR STRUCTURE OR INSIDE NEAREST POINT OF ENTRANCE OF THE SYSTEM CONDUCTORS.
 - HEIGHT OF THE INTEGRATED AC/DC DISCONNECT SHALL NOT EXCEED 6'-7" PER NEC CODE 240.24.
 - A GROUNDING ELECTRODE SYSTEM IN ACCORDANCE WITH NEC 690.47 AND 250.50 THROUGH 60 AND 250-166 SHALL BE PROVIDED. PER NEC
 - GROUNDING ELECTRODE SYSTEM OF EXISTING BUILDING MAY BE USED AND BONDED TO THE SERVICE ENTRANCE. IF EXISTING SYSTEM IS INACCESSIBLE OR INADEQUATE A SUPPLEMENTAL GROUNDING ELECTRODE WILL BE USED AT THE INVERTER LOCATION CONSISTING OF A UL LISTED 8 FT. GROUND ROD WITH ACORN CLAMP. GROUNDING ELECTRODE CONDUCTORS SHALL BE NO LESS THAN #8 AWG AND NO LARGER THAN #6 AWG COPPER AND BONDED TO THE EXISTING GROUNDING ELECTRODE TO PROVIDE FOR A COMPLETE SYSTEM.
 - PHOTOVOLTAIC MODULES ARE TO BE CONSIDERED NON-COMBUSTIBLE.
 - PHOTOVOLTAIC INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING, MECHANICAL, OR BUILDING ROOF VENTS.
 - WATER PROOF CONNECTORS AND HUBS SHALL BE USED WHERE APPLICABLE PER NEC 312.2 AND 314.15.
 - GROUNDING/BONDING BUSHINGS SHALL BE INSTALLED WHERE APPLICABLE PER NEC 250.92.
 - ALL EXTERIOR RACEWAYS ON WALLS SHOULD BE EMT UNLESS NOTED OTHERWISE.
 - CONTRACTOR SHALL INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS & NOTHING IN THIS DOCUMENT SHALL BE INTERPRETED IN A WAY THAT OVERRIDES THEM.
 - CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL CONDITIONS, DIMENSIONS & DETAILS IN THIS DOCUMENT. ALL DIMENSIONS SHALL BE VERIFIED BY SUBCONTRACTOR BEFORE INSTALLATION.
- PHOTOVOLTAIC SYSTEM FIRE CLASSIFICATION LISTING IN ACCORDANCE WITH UL1703 STANDARD.

CONTRACTOR INFO:

DRAWN DATE	07/07/2025
PROJECT ID	SP - 1
CREATED BY	RK
CHECKED BY	MSB
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE

1234 STREET NAME, CITY, NY 9874

APN: ----

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-0

SHEET NO.

PROJECT SUMMARY

SHEET INFORMATION

ELECTRICAL NOTES

1.

1.INSTALLATION TO BE COMPLIANT WITH NATIONAL FIRE CODE (NFC) ARTICLE NFPA 1 & NFPA70
2.

THE PV SOLAR SYSTEM IS AN UNGROUNDED PV SOLAR SYSTEM ARRAY AND HAS A GROUND-FAULT PROTECTION DEVICE THAT MEETS THE REQUIREMENTS OF NATIONAL ELECTRICAL CODE (NEC) ARTICLE 690.41(B)(1) AND (2)
3.

THE EXACT LOCATION OF NEW ELECTRICAL EQUIPMENT AND CONDUIT RUN RELATING TO THIS PROJECT IS SUBJECT TO CHANGE AND WILL BE DETERMINED ON-SITE BY THE CONTRACTOR.
4.

ALL CLEARANCES AND WORKSPACE, AS REQUIRED PER NEC 110.26, SHALL BE FOLLOWED
5.

THE INVERTER(S) SHALL MEET ALL CURRENT CODE REQUIREMENTS FOR RAPID SHUTDOWN (RSD) AS DEFINED IN NEC 690.12.
6.

ALL EQUIPMENT TO BE LISTED OR LABELED FOR ITS APPLICATION (UL OR OTHER APPROVED LISTINGS)

i.

PV SOLAR MODULE - UL 1703

ii.

INVERTER - UL 1741

iii.

PV SOLAR MOUNTING SYSTEM - UL 2703
7.

GROUNDING

i.

ALL EQUIPMENT SHALL BE GROUNDED PER NEC ARTICLES 250 & 690 7.2 REQUIREMENTS.

ii.

MODULE BONDING METHOD SHALL BE INTEGRATED GROUNDING MID-CLAMPS. REFER TO MANUFACTURES SPECIFIC INSTRUCTIONS FOR PROPER BONDING TECHNIQUES.

iii.

GROUNDING SYSTEM COMPONENTS SHALL BE LISTED FOR THEIR PURPOSE, AND GROUNDING DEVICES SHALL BE RATED FOR DIRECT BURIAL

iv.

EQUIPMENT GROUNDING CONNECTOR (EGC) SHALL BE SIZED FOLLOWING NEC 250.122, AND ARRAY EGCS SMALLER THAN 6 AWG SHALL COMPLY WITH NEC 250.120(C)
8.

ALL CONDUCTORS ARE COPPER UNLESS SPECIFIED OTHERWISE
9.

ALL CONDUIT, RACEWAYS, AND JUNCTION BOXES SHALL BE SIZED ACCORDING TO THE APPLICABLE CODE IF THE SIZE IS NOT SPECIFIED.
10.

SIGNAGE SHALL BE APPLIED ACCORDING TO GOVERNING BUILDING CODES AND LOCAL JURISDICTIONS' SPECIFIC REQUIREMENTS.
11.

EQUIPMENT INSTALLED IN DIRECT SUNLIGHT MUST BE RATED FOR EXPECTED OPERATING TEMPERATURE AS SPECIFIED BY NEC.
12.

CALCULATION OF MAXIMUM CIRCUIT CURRENT FOR THE SPECIFIC CIRCUIT SHALL BE CALCULATED BY NEC ARTICLE 690.8(A)(1) THROUGH (A)(6). CONDUCTOR AMPACITY SHALL BE SIZED NOT TO CARRY LESS THAN THE LARGER OF NEC ARTICLE 690.8(B)(1) OR (2)
13.

DC PV SOLAR SYSTEM SOURCE AND DC OUTPUT CURRENT CIRCUITS ON OR INSIDE A BUILDING SHALL BE CONTAINED IN METAL RACEWAYS, TYPE MC METAL-CLAD CABLE THAT COMPLIES WITH NEC SECTION 250.118(10), OR METAL ENCLOSURES FROM THE POINT OF PENETRATION OF THE SURFACE OF THE BUILDING OR STRUCTURE TO THE FIRST READILY ACCESSIBLE DISCONNECTING MEANS AS DESCRIBED IN NEC 690.31(G)
14.

ENSURE THAT JUNCTION, PULL, AND OUTLET BOXES LOCATED BEHIND MODULES OR PANELS ARE INSTALLED IN A WAY THAT ALLOWS THE WIRING INSIDE TO BE EASILY ACCESSED EITHER DIRECTLY OR BY MOVING A SECURED MODULE(S) WITH REMOVABLE FASTENERS AND CONNECTED BY A FLEXIBLE WIRING SYSTEM AS DESCRIBED IN NEC 690.34
15.

PV SOLAR SYSTEM POINT OF CONNECTION (POC). THE OUTPUT OF AN INTERCONNECTED ELECTRIC POWER SOURCE SHALL BE CONNECTED AS SPECIFIED IN NEC ARTICLE 705.12(A) OR (B).

CONTRACTOR INFO:

DRAWN DATE	07/07/2025
PROJECT ID	SP - 1
CREATED BY	RK
CHECKED BY	MSB
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

CLIENT RESIDENCE

1234 STREET NAME, CITY, NY 9874

APN: ----

#	REVISION/UPDATE	DATE

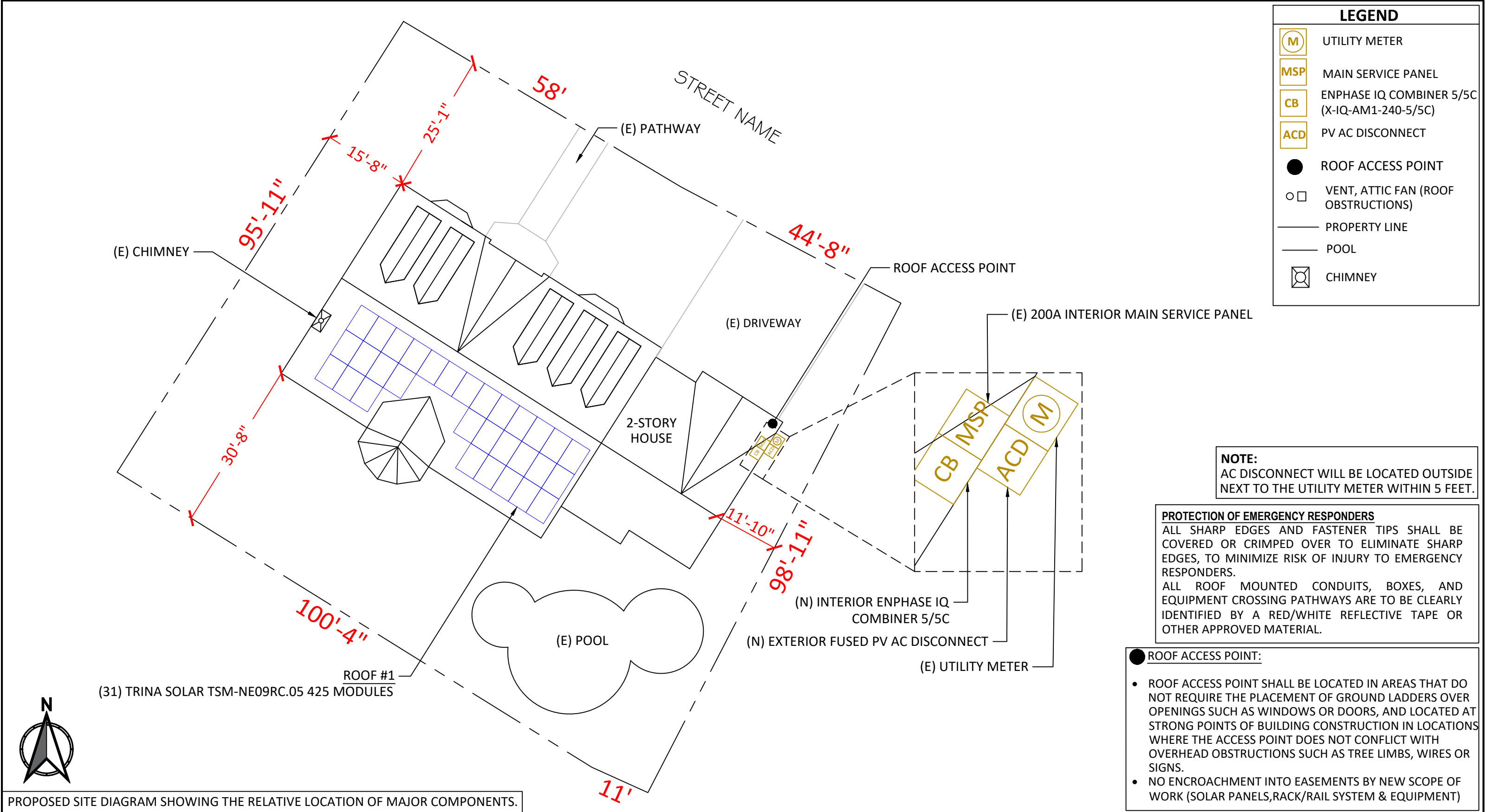
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PV-0.1

SHEET NO.

PROJECT SUMMARY

SHEET INFORMATION



CONTRACTOR INFO:	
------------------	--

DRAWN DATE	07/07/2025
PROJECT ID	SP - 1
CREATED BY	RK
CHECKED BY	MSB
SCALE	1/16" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO	
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CLIENT RESIDENCE
1234 STREET NAME, CITY, NY 9874
APN: ----

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-1 SHEET NO.
SITE PLAN
SHEET INFORMATION

ROOF DETAILS	
MOUNTING PLANE	ROOF #1
NO. OF MODULES	31
AZIMUTH	212°
ROOF TILT	26°
ROOF TYPE	COMP. SHINGLE
RAFTER SIZE	2"X12"
RAFTER SPACING	16" O.C
ARRAY AREA	666.80
(TOTAL ARRAY AREA/ TOTAL ROOF AREA) X 100% = (666.80/3059.07) X 100% = 21.80%	

NOTE: ROOF ACCESS, PATHWAYS AND SPACING REQUIREMENTS SHALL BE PROVIDED IN ACCORDANCE WITH SECTIONS 1205.2.1 THROUGH 1205.3.3. PATHWAYS SHALL BE OVER AREAS CAPABLE OF SUPPORTING FIRE FIGHTERS ACCESSING THE ROOF. PATHWAYS SHALL BE LOCATED IN AREAS WITH MINIMAL OBSTRUCTIONS, SUCH AS VENT PIPES, CONDUIT OR MECHANICAL EQUIPMENT.

LEGEND

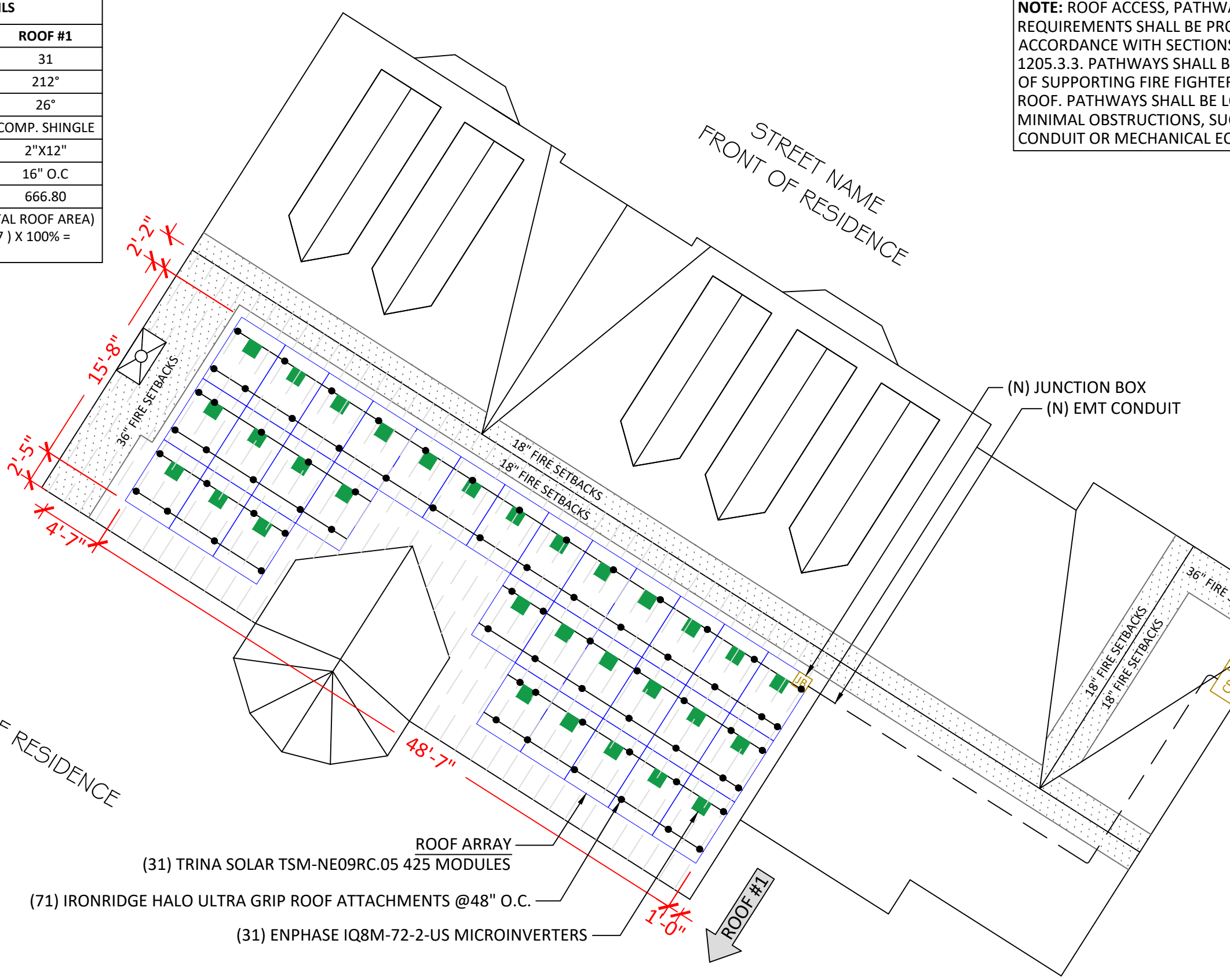
- M

UTILITY METER
- MSP

MAIN SERVICE PANEL
- CB

ENPHASE IQ COMBINER 5/5C (X-IQ-AM1-240-5/5C)
- ACD

PV AC DISCONNECT
- ENPHASE IQ8M-72-2-US MICROINVERTER
- VENT, ATTIC FAN (ROOF OBSTRUCTIONS)
- IRONRIDGE HALO ULTRA GRIP ATTACHMENT
- RAFTER
- IRONRIDGE AIRE A1 FLUSH MOUNT
- EMT CONDUIT
- POOL
- CHIMNEY



NOTES: ATTACHMENT POINTS ARE NOT EXACT. CONTRACTOR MAY NEED TO ADJUST MOUNT LOCATION. EXISTING ROOF CONDITION & RAFTER LOCATIONS ARE SUBJECT TO FIELD VERIFICATION. ACTUAL LOCATION MAY DIFFER. INSTALL PER MANUFACTURER SPECIFICATIONS. SEE SUPPLEMENTAL DOCUMENTS FOR ATTACHMENT & STRUCTURAL DOCUMENTATION.



CONTRACTOR INFO:

DRAWN DATE	07/07/2025
PROJECT ID	SP - 1
CREATED BY	RK
CHECKED BY	MSB
SCALE	1/8" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

CLIENT RESIDENCE

1234 STREET NAME, CITY, NY 9874

APN: ----

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-2

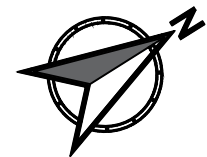
SHEET NO.

ARRAY ATTACHMENT DETAILS

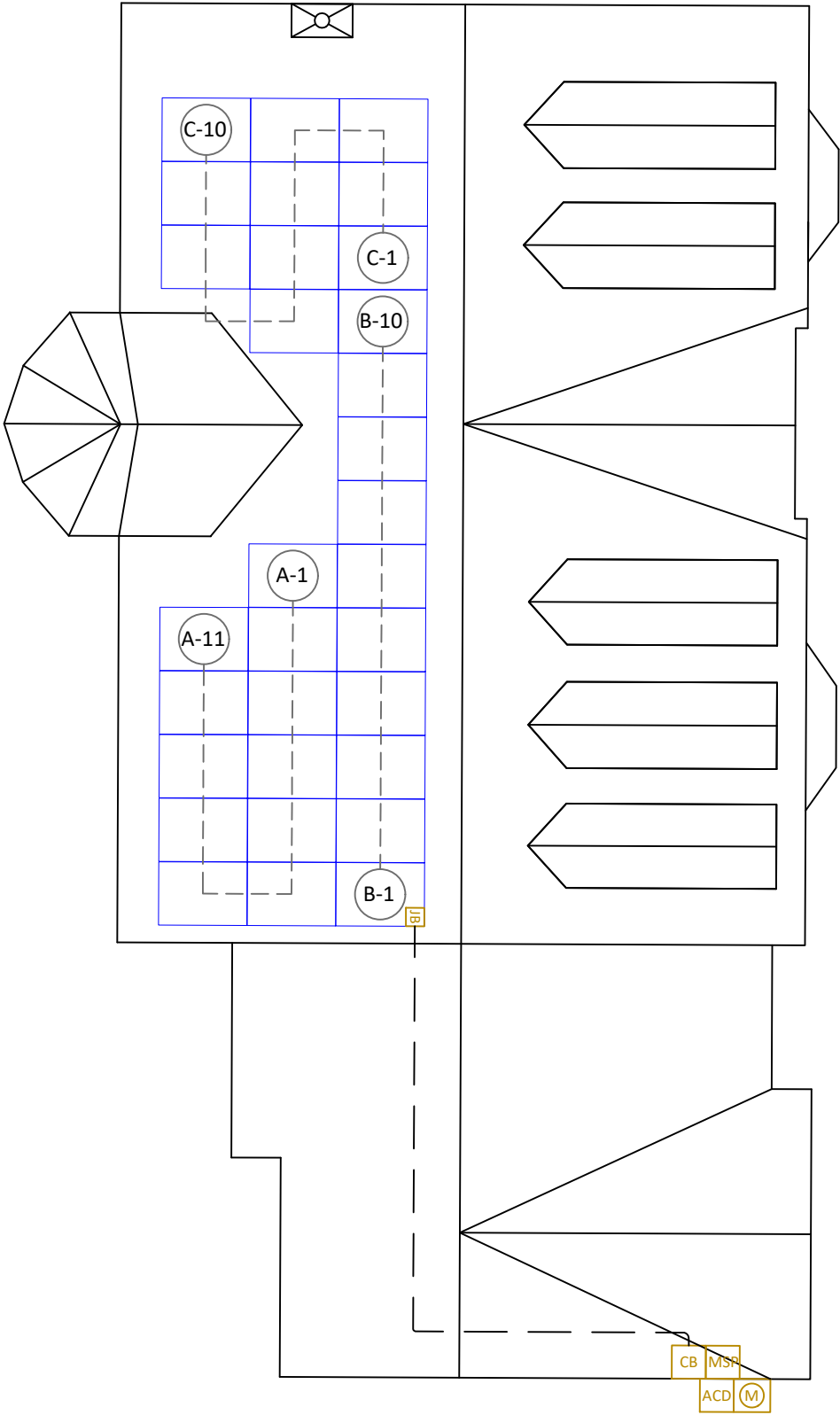
SHEET INFORMATION

BILL OF MATERIALS		
EQUIPMENT	QTY	DESCRIPTION
SOLAR PV MODULE	31	TRINA SOLAR TSM-NE09RC.05 425 MODULES
MICRO-INVERTER	31	ENPHASE IQ8M-72-2-US MICRO-INVERTERS
JUNCTION BOX	1	JUNCTION BOX, 600V, NEMA 3R, UL LISTED
COMBINER BOX	1	ENPHASE IQ COMBINER 5/5C (X-IQ-AM1-240-5/5C)
PV AC DISCONNECT	1	60A FUSED PV AC DISCONNECT, (2) 60A FUSES, 240V, NEMA 3R,UL LISTED
ENPHASE Q CABLE	35	ENPHASE Q CABLE 240V, (PER CONNECTOR)
BRANCH TERMINATOR	3	BRANCH TERMINATOR
WATER TIGHT CAPS	4	WATER TIGHT CAPS
ATTACHMENT	71	IRONRIDGE HALO ULTRA GRIP
ATTACHMENT	71	FLASHING, ROUNDED CORNER, 9"x12"
ATTACHMENT	71	LAG SCREW, HEX HEAD, 14"x3"
ATTACHMENT	71	14"x3" PER HALO
RAIL	17	IRONRIDGE AIRE A1 FLUSH MOUNT (170")
BONDED SPLICE	12	BONDED SPLICE
CLAMPS	72	UNIVERSAL FASTENING OBJECTS (UFO)
CLAMPS	20	STOPPER SLEEVE
GROUNDING LUG	5	GROUNDING LUG

ARRAY WEIGHT (LOAD CALC'S)		
Number of Modules	31	
Module Weight	48	LBS
Total Module (Array) Weight	1489.86	LBS
Number of Attachment point	71	
Mounting System Weight	1	LBS
(Per Module)		
Mounting System Weight	71.00	LBS
Total System Weight	1560.86	LBS
(Module Weight + Mounting System Weight)		
Weight at Each Attachment Point	20.98	LBS
(Array Weight / Number of Attachment Point)		
Module Area (69.37"x44.65")	21.51	SqFt
Total Array Area	666.80	SqFt
Distributed Load	2.34	Per SqFt
(Total System Weight / Total Array Area)		
Total Roof Area	3059.07	SqFt
Total Percentage of Roof Covered	21.80%	
Total Array Area / Total Roof Area)*100		



BACK OF RESIDENCE



STREET NAME
FRONT OF RESIDENCE

LEGEND	
	UTILITY METER
	MAIN SERVICE PANEL
	ENPHASE IQ COMBINER 5/5C (X-IQ-AM1-240-5/5C)
	PV AC DISCONNECT
	JUNCTION BOX
	VENT, ATTIC FAN (ROOF OBSTRUCTIONS)
	EMT CONDUIT
	POOL
	CHIMNEY

BRANCH LEGEND	
	BRANCH #1 - 11 MODULES
	BRANCH #2 - 10 MODULES
	BRANCH #3 - 10 MODULES

CONTRACTOR INFO:

DRAWN DATE	07/07/2025
PROJECT ID	SP - 1
CREATED BY	RK
CHECKED BY	MSB
SCALE	1/10" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE

1234 STREET NAME, CITY, NY 9874

APN: ----

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-3

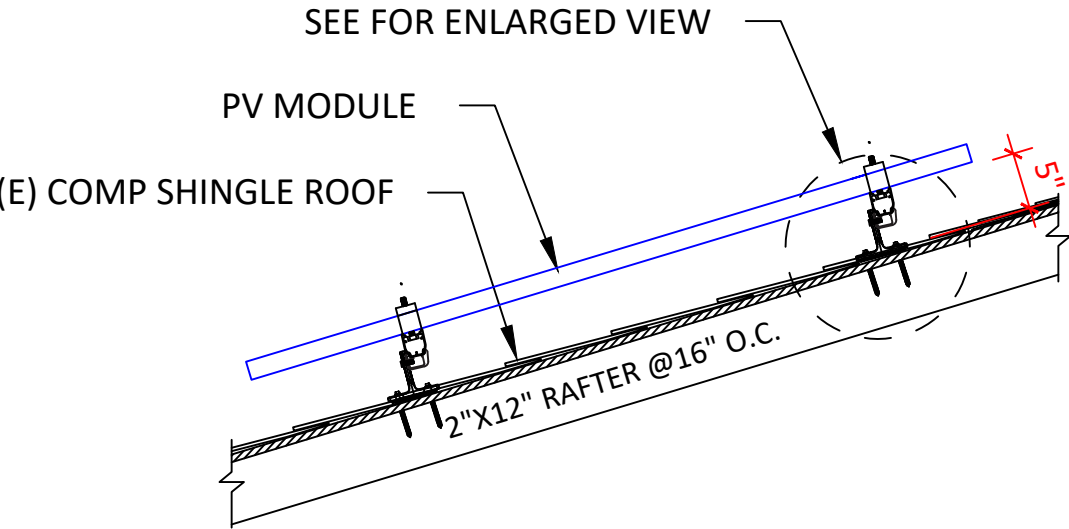
SHEET NO.

BOM & STRING LAYOUT

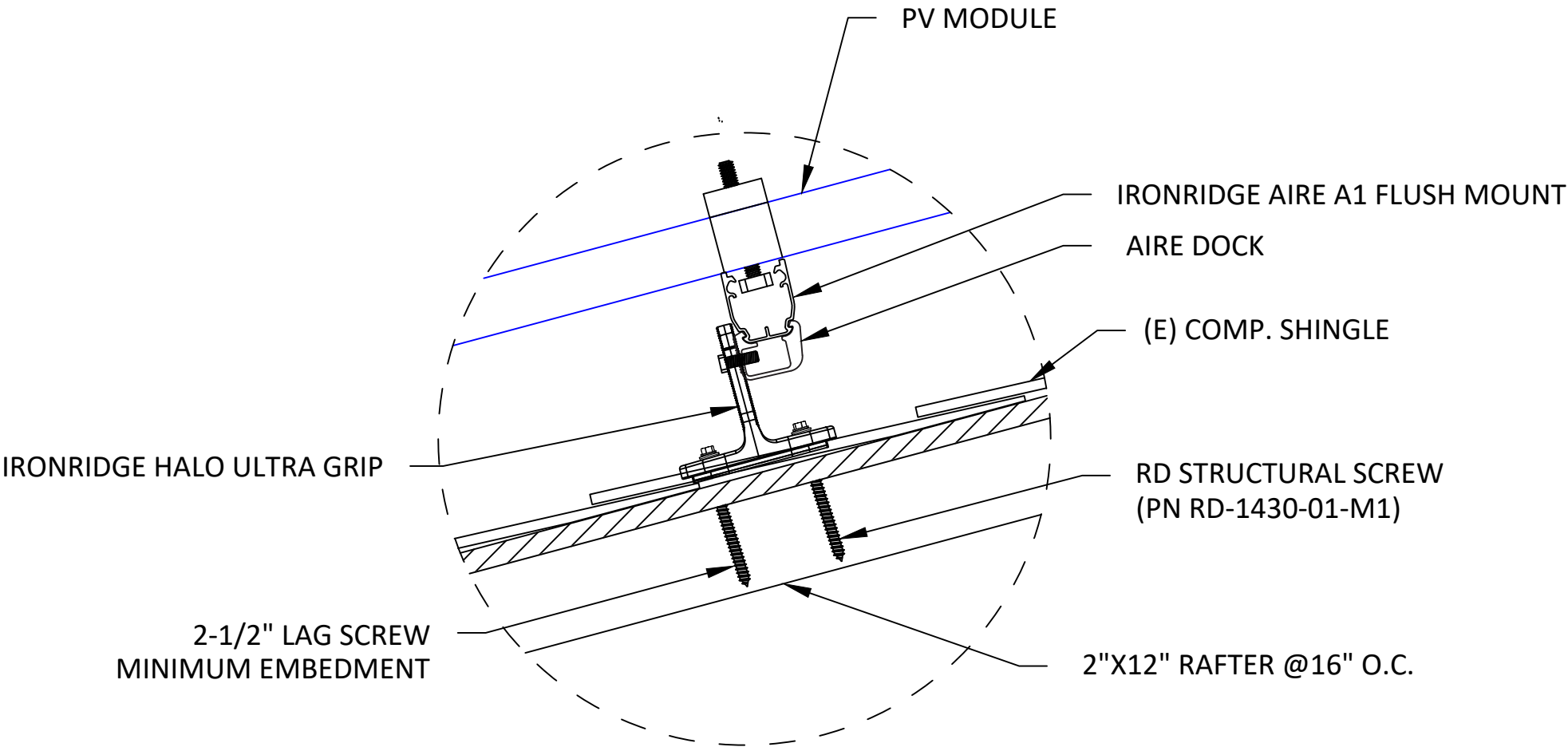
SHEET INFORMATION

STRUCTURAL NOTES:

- 1. THE SOLAR PANELS ARE TO BE MOUNTED TO THE ROOF FRAMING USING THE IRONRIDGE AIRE A1 FLUSH MOUNT SYSTEM WITH IRONRIDGE HALO ULTRA GRIP ATTACHMENTS.
- 2. THE MOUNTING FEET ARE TO BE SPACED AS SHOWN IN THE DETAILS, AND MUST BE STAGGERED TO ADJACENT FRAMING MEMBERS TO SPREAD OUT THE ADDITIONAL LOAD.
- 3. THE PROPOSED PV SYSTEM ADDS 2.34 PSF TO THE ROOF FRAMING SYSTEM.
- 4. ROOF LIVE LOAD = 20 PSF
- 5. GROUND SNOW LOAD = 25 lb/sqft
- 6. WIND SPEED = 125 MPH
- 7. EXPOSURE CATEGORY = B
- 8. RISK CATEGORY = II



STRUCTURAL ELEVATION DIAGRAM



ATTACHMENT DETAILED DIAGRAM

NOTES: FLASHING SHALL BE APPLIED IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS. IF THERE IS ANY CONFLICT BETWEEN WHAT IS DEPICTED & INSTRUCTION AS PER MANUFACTURER, MANUFACTURERS INSTRUCTIONS SHALL SUPERSEDE.

CONTRACTOR INFO:	DRAWN DATE	07/07/2025
	PROJECT ID	SP - 1
	CREATED BY	RK
	CHECKED BY	MSB
	SCALE	NTS
	SHEET SIZE	11"X 17"
		ANSI B

HOMEOWNER INFO	CLIENT RESIDENCE	
	1234 STREET NAME, CITY, NY 9874	
	APN: ----	

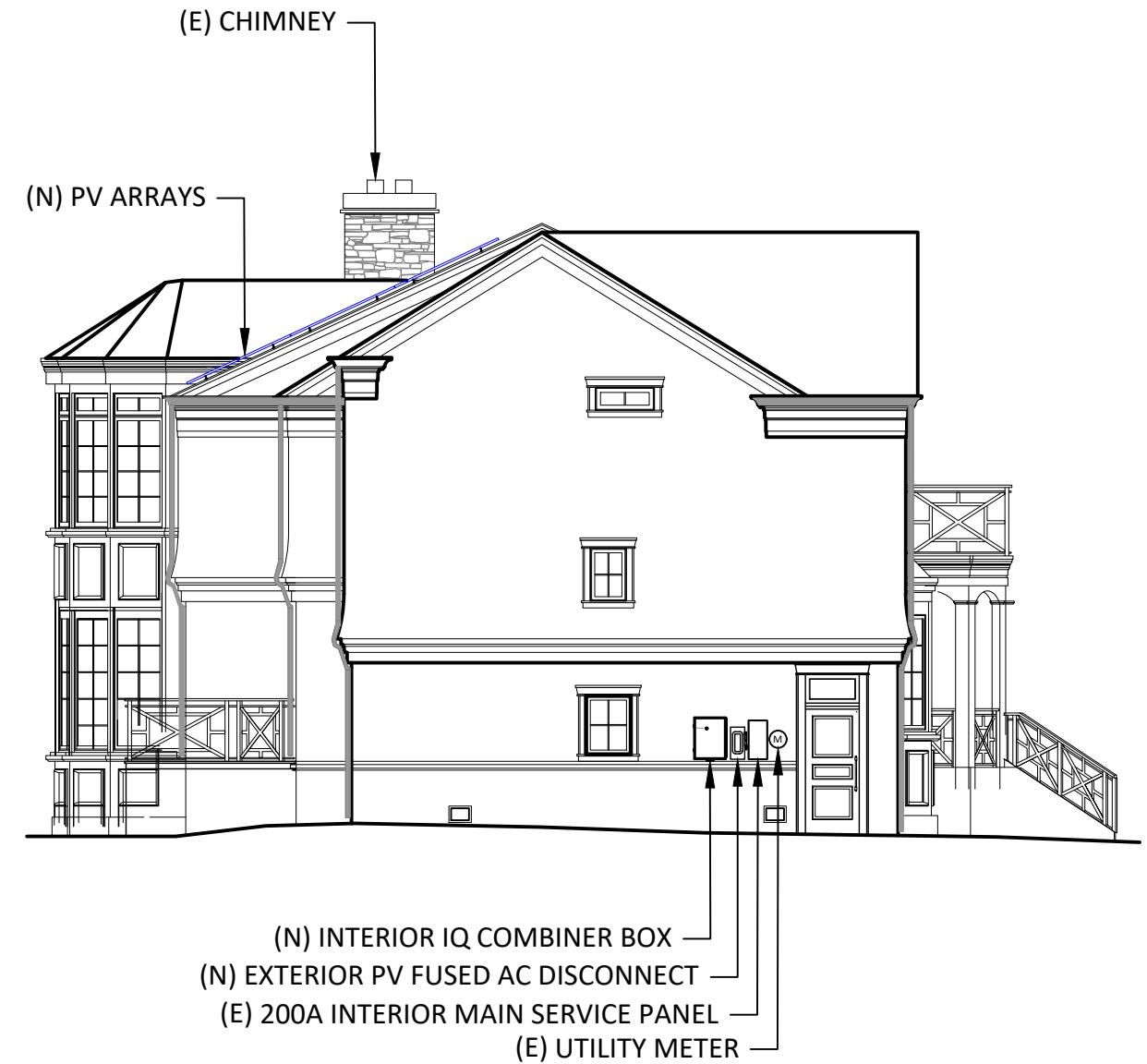
#	REVISION/UPDATE	DATE

SIGNED & STAMP BY	

SHEET INFORMATION	S-1
	SHEET NO.
	STRUCTURAL DETAILED DIAGRAM



FRONT SIDE OF HOUSE



LEFT SIDE OF HOUSE

CONTRACTOR INFO:	DRAWN DATE	07/07/2025	CLIENT RESIDENCE 1234 STREET NAME, CITY, NY 9874 APN: ----	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	S-2 SHEET NO. EQUIPMENT ELEVATION DIAGRAM SHEET INFORMATION
	PROJECT ID	SP - 1						
	CREATED BY	RK						
	CHECKED BY	MSB						
	SCALE	NTS						
	SHEET SIZE	11"X 17"	HOMEOWNER INFO					
		ANSI B						

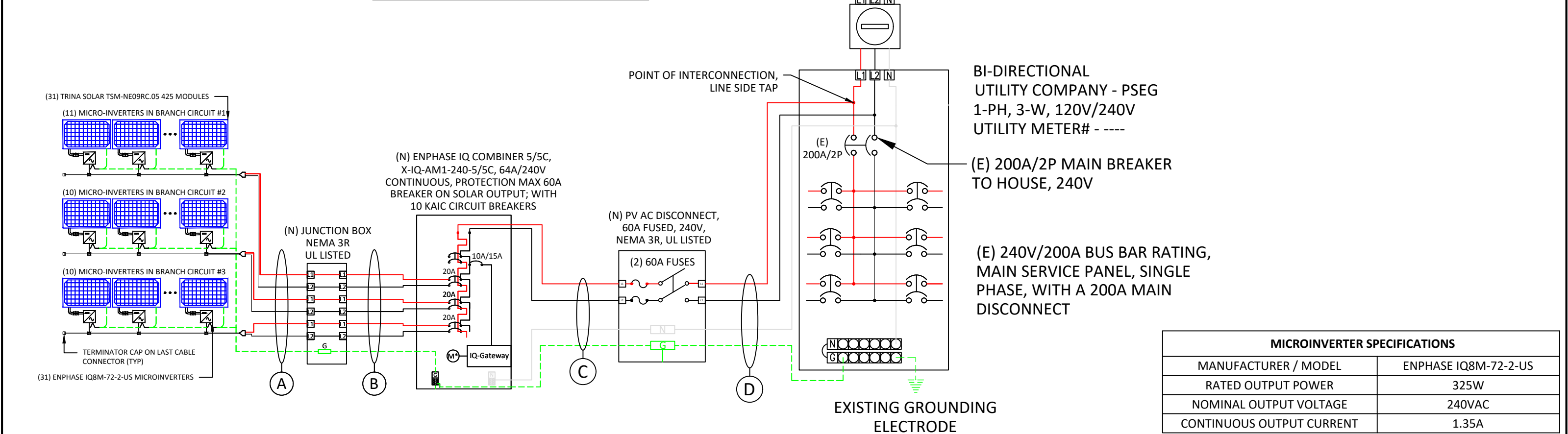
[DC] SYSTEM SIZE		[AC] SYSTEM SIZE	
(31) MODULES x 425W		(31) MICROINVERTERS x 325W	
13.18 kW		10.08 kW	

SOLAR MODULE SPECIFICATIONS	
MANUFACTURER / MODEL	TRINA SOLAR TSM-NE09RC.05 425
VMP	43.2V
IMP	9.84A
VOC	51.4V
ISC	10.47A
TEMP. COEFF. VOC	-0.25%/°C
MODULE DIMENSION	69.37" (L) x 44.65" (W)
PANEL WATTAGE	425W

NOTE:
AC DISCONNECT WILL BE LOCATED OUTSIDE
NEXT TO THE UTILITY METER WITHIN 5 FEET.

AMBIENT TEMPERATURE SPECS	
RECORD LOW TEMP	-14°C
AMBIENT TEMP (HIGH TEMP 2%)	31°C
CONDUIT HEIGHT	7/8"
ROOF TOP TEMP	90°C
CONDUCTOR TEMPERATURE RATE	53°C
MODULE TEMPERATURE COEFFICIENT OF VOC	-0.25%/°C

EQUIPMENT DETAILS	
(#) PV MODULES	(31) TRINA SOLAR TSM-NE09RC.05 425
(#) MICROINVERTER	(31) ENPHASE IQ8M-72-2-US
(#) COMBINER BOX	(01) ENPHASE IQ COMBINER 5 (X-IQ-AM1-240-5)
OUTPUT CURRENT	41.85A
BRANCH CIRCUIT	2 BRANCH CIRCUITS OF 10 MODULES 1 BRANCH CIRCUIT OF 11 MODULES
POINT OF INTERCONNECTION	LINE SIDE TAP IN THE INTERIOR MAIN SERVICE PANEL WITH 200A MAIN DISCONNECT, 200A BUSBAR, 120/240V, 3W, 1φ
UTILITY AHJ	PSEG TOWN OF HEMPSTEAD



ID	CONDUCTOR			CONDUIT	# OF PARALLEL CIRCUITS	CURRENT-CARRYING CONDUCTORS IN CONDUIT	CONDUIT FILL PERCENT	OCPD	EGC		TEMP. CORR. FACTOR		CONDUIT FILL FACTOR	CONT. CURRENT (INV. CONTINUOUS CURRENT x 1.25)				MAX CURRENT	BASE AMP.	DERATED AMP.	LENGTH	VOLTAGE DROP
A	12 AWG	Q CABLE		-	1	2	N/A	N/A	6 AWG	BARE COPPER	0.76	(53°C)	N/A	14.8A	x	1.25	=	18.5A	N/A	N/A	36FT	0.62%
B	10 AWG	THWN-2	COPPER	MIN 0.75" Dia EMT	3	6	32.04%	20A	6 AWG	THWN-2, COPPER	0.94	(31°C)	0.8	14.8A	x	1.25	=	18.5A	35A	35A x 0.8 x 0.94 = 26.32A	61FT	0.97%
C	6 AWG	THHN	COPPER	MIN 0.75" Dia EMT	1	3	36.53%	60A	6 AWG	THWN, COPPER	0.94	(31°C)	1	41.8A	x	1.25	=	52.2A	65A	65A x 1 x 0.94 = 61.1A	5FT	0.10%
D	6 AWG	THHN	COPPER	MIN 0.75" Dia EMT	1	3	36.53%	N/A	6 AWG	THWN, COPPER	0.94	(31°C)	1	41.8A	x	1.25	=	52.2A	65A	65A x 1 x 0.94= 61.1A	5FT	0.10%

CONTRACTOR INFO:	DRAWN DATE	07/07/2025	CLIENT RESIDENCE 1234 STREET NAME, CITY, NY 9874 APN: ----	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	E-1 SHEET NO. ELECTRICAL DIAGRAM & CALCULATIONS SHEET INFORMATION
	PROJECT ID	SP - 1						
	CREATED BY	RK						
	CHECKED BY	MSB						
	SCALE	NTS						
HOMEOWNER INFO:	SHEET SIZE	11"X 17"						
		ANSI B						

1

!

WARNING

ELECTRIC SHOCK HAZARD

DO NOT TOUCH TERMINALS.
TERMINALS ON BOTH LINE AND LOAD SIDES MAY
BE ENERGIZED IN THE OPEN POSITION

LABEL LOCATION:
MAIN SERVICE PANEL/AC DISCONNECT/INVERTER/
AC COMBINER
2017 NEC 690.13(B)

2

!

WARNING

ELECTRIC SHOCK HAZARD

IF GROUND FAULT IS INDICATED
ALL NORMALLY GROUNDED
CONDUCTORS MAY BE
UNGROUND AND ENERGIZED

LABEL LOCATION:
INVERTER(S), ENPHASE ENVOY ENCLOSURE (IF
APPLICABLE).
PER CODE(S): NEC 2017: 690.5(C)

3

PHOTOVOLTAIC AC DISCONNECT

RATED AC OUTPUT CURRENT 41.85A

NOMINAL OPERATING AC VOLTAGE 240 VAC

LABEL LOCATION:
MAIN SERVICE PANEL/AC DISCONNECT
2017 NEC 690.53

4

RAPID SHUTDOWN SWITCH
FOR SOLAR PV SYSTEM

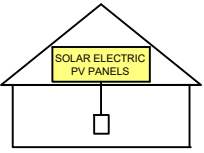
CONFIRM AND LABEL EXACT RSD SWITCH LOCATION:
MICRO-INVERTER
AT OR WITHIN 3' OF THE DC COMBINER
SWITCH
2017 NEC 690.56(C)(3)

ADHESIVE FASTENED SIGNS:
• LABELS SHALL BE MADE OF RED PLASTIC MATERIAL WITH ENGRAVED WHITE LETTERS.
• LETTERS SHALL BE A MINIMUM 3/8" IN SIZE.
• THE LABELS SHALL BE PERMANENTLY ATTACHED TO THE APPROPRIATE PANEL.
• AC & DC CONDUIT, RACEWAY, ENCLOSURES, CABLE ASSEMBLIES AND JUNCTION BOXES SHALL BE RED BACKGROUND MATERIAL WITH WHITE LETTERING MADE OF DURABLE ADHESIVE, REFLECTIVE, WEATHER RESISTANT MATERIAL SUITABLE FOR THE ENVIRONMENT PER UL 969; TO ALERT FIRE SERVICE TO AVOID CUTTING THEM OFF.

5

SOLAR PV SYSTEM EQUIPPED
WITH RAPID SHUTDOWN

TURN RAPID SHUTDOWN
SWITCH TO THE
"OFF" POSITION TO
SHUT DOWN PV SYSTEM
AND REDUCE
SHOCK HAZARD
IN THE ARRAY



LABEL LOCATION:
MAIN SERVICE PANEL
IF MSD IS OUTSIDE PLACE IT THERE / IF
MSD IS INSIDE PLACE ON THE AC DISCONNECT
2017 NEC 690.56(C)

6

!

WARNING

PHOTOVOLTAIC SYSTEM
COMBINER PANEL

DO NOT ADD LOADS

LABEL LOCATION:
PHOTOVOLTAIC AC COMBINER (IF APPLICABLE).
PER CODE(S): NEC 2017: 705.12(B)(3)(2)

7

PHOTOVOLTAIC SYSTEM
UTILITY DISCONNECT SWITCH

LABEL LOCATION:
AC DISCONNECT
2017 NEC 690.56(C)(3)

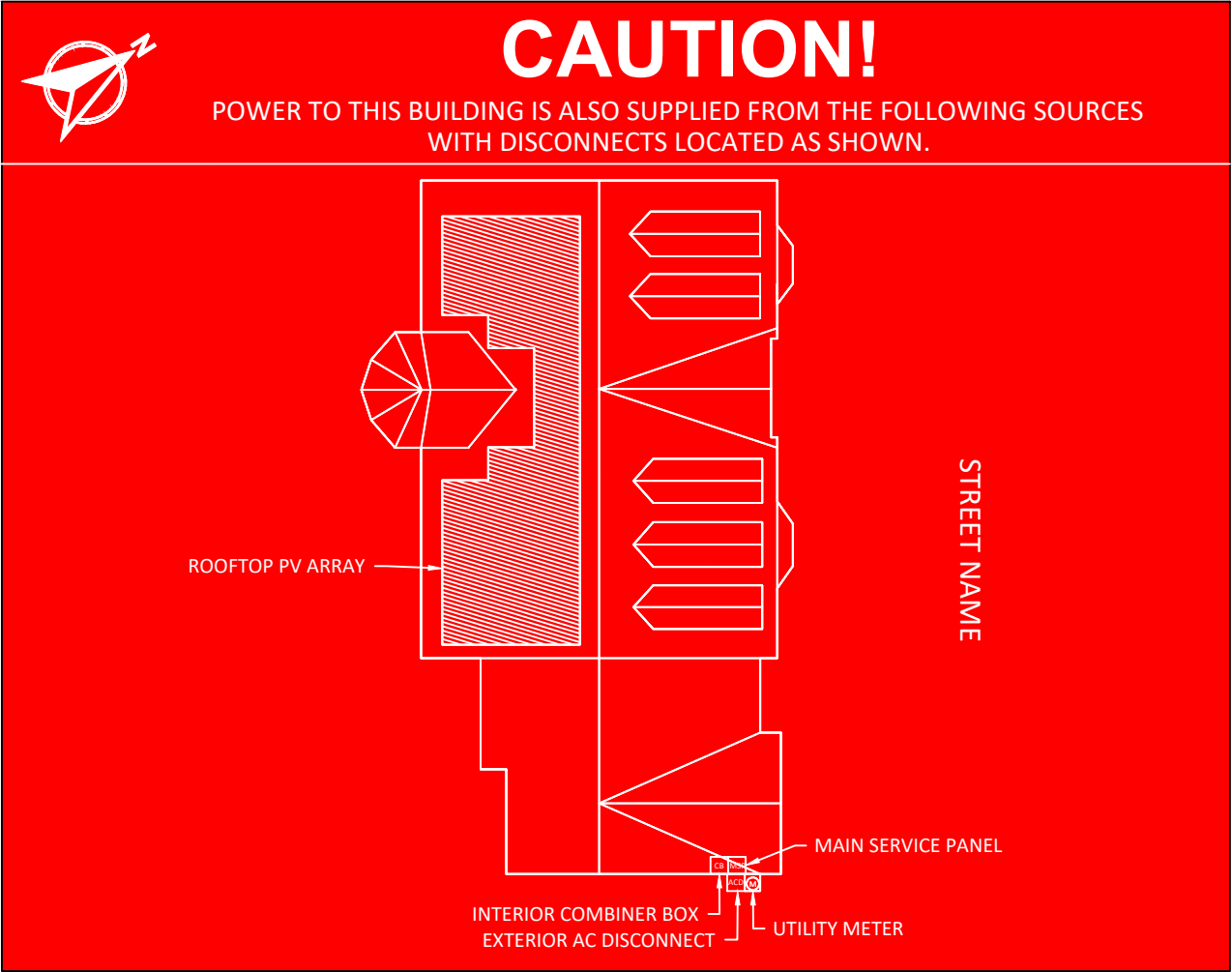
8

!

CAUTION

DUAL POWER SOURCE
SECOND SOURCE IS
PHOTOVOLTAIC

LABEL LOCATION:
MAIN SERVICE PANEL/AC DISCONNECT/AC COMBINER/
REVENUE METER
2017 NEC 705.12(D)(3)



LABEL LOCATION:
EACH SERVICE EQUIPMENT LOCATION AND AT THE LOCATION(S) OF THE SYSTEM DISCONNECT(S) FOR ALL ELECTRIC POWER
PRODUCTION SOURCES CAPABLE OF BEING INTERCONNECTED

CONTRACTOR INFO:

DRAWN DATE	07/07/2025
PROJECT ID	SP - 1
CREATED BY	RK
CHECKED BY	MSB
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE

1234 STREET NAME, CITY, NY 9874

APN: ----

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

E-2

SHEET NO.

PLACARDS

SHEET INFORMATION



N-TYPE MODULE

425W

MAXIMUM POWER OUTPUT

0~+5W

POSITIVE POWER TOLERANCE

21.3%

MAXIMUM EFFICIENCY

PRODUCT: TSM-NE09RC.05

PRODUCT RANGE: 400W -425W



Smaller panel, Bigger power generation

- Up to 425W, 21.3% module efficiency.
- Reduce installation cost with higher module power on the roof.
- Boost performance in warm weather with lower temperature coefficient



High Reliability

- Innovative non-destructive cell cutting technology for improved mechanical resistance and strength



Lower Degradation, longer warranty, higher output

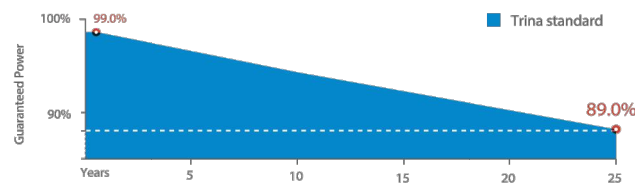
- First-year degradation 1% and annual degradation at 0.4%
- Up to 25 years product warranty and 25 years power warranty



Universal solution for residential and C&I rooftops

- Easy for integration, designed for compatibility with existing mainstream inverters and diverse mounting systems
- Perfect size and low weight for handling and installation
- Most valuable solution on low load capacity rooftops
- Mechanical performance up to 6000 Pa positive load and 4000 Pa negative load

Trina Solar's Vertex Bifacial Dual Glass Performance Warranty



Comprehensive Products and System Certificates

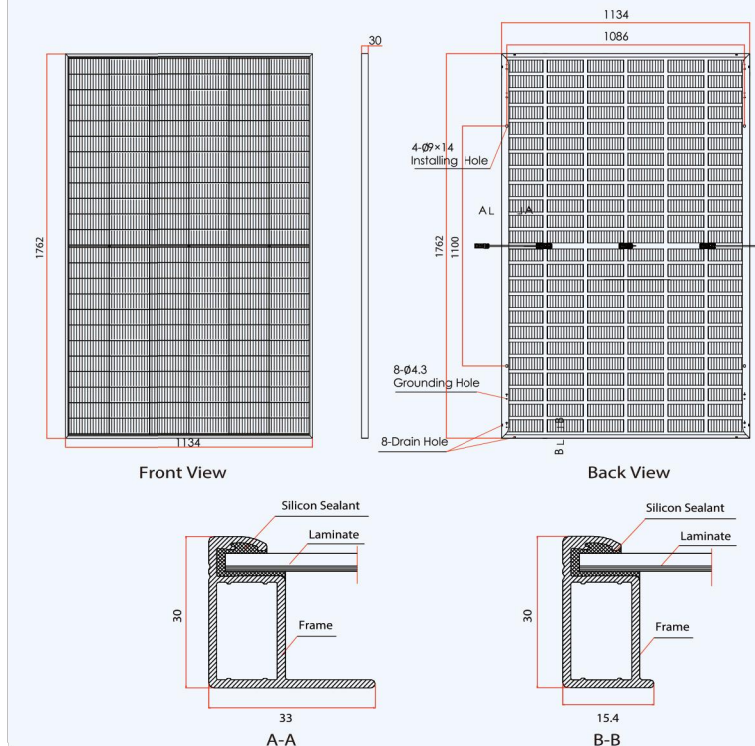


IEC61215/IEC61730/IEC61701/IEC62716/UL61730
ISO 9001: Quality Management System
ISO 14001: Environmental Management System
ISO14064: Greenhouse Gases Emissions Verification
ISO45001: Occupational Health and Safety Management System

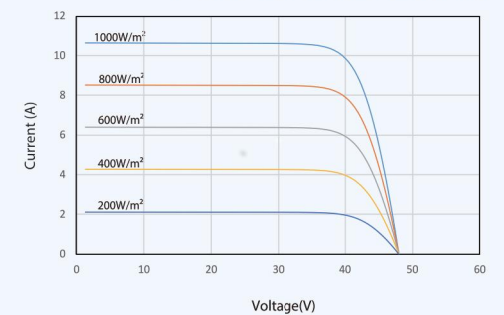


N-TYPE MODULE

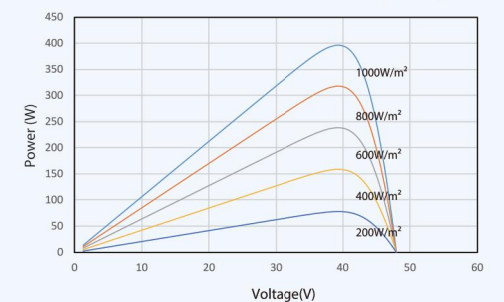
DIMENSIONS OF PV MODULE(mm)



I-V CURVES OF PV MODULE(420 W)



P-V CURVES OF PV MODULE(420W)



ELECTRICAL DATA (STC)

Peak Power Watts-P _{MAX} (Wp)*	400	405	410	415	420	425
Power Tolerance-P _{MAX} (W)	0 ~ +5					
Maximum Power Voltage-V _{MPP} (V)	41.3	41.7	42.1	42.5	42.8	43.2
Maximum Power Current-I _{MPP} (A)	9.68	9.71	9.73	9.77	9.80	9.84
Open Circuit Voltage-V _{OC} (V)	49.2	49.6	50.1	50.5	50.9	51.4
Short Circuit Current-I _{SC} (A)	10.30	10.33	10.37	10.40	10.43	10.47
Module Efficiency η _m (%)	20.0	20.3	20.5	20.8	21.0	21.3

STC: Irradiance 1000W/m², Cell Temperature 25°C, Air Mass AM1.5.

*Measuring tolerance: ±3%.

Electrical characteristics with different power bin (reference to 10% Irradiance ratio)

Total Equivalent power -P _{MAX} (Wp)	426	431	437	442	447	453
Maximum Power Voltage-V _{MPP} (V)	41.3	41.7	42.1	42.5	42.8	43.2
Maximum Power Current-I _{MPP} (A)	10.31	10.34	10.36	10.41	10.44	10.48
Open Circuit Voltage-V _{OC} (V)	49.2	49.6	50.1	50.5	50.9	51.4
Short Circuit Current-I _{SC} (A)	10.97	11.00	11.04	11.08	11.11	11.15
Irradiance ratio (rear/front)	10%					

Power Bifaciality: ±5±10%.

ELECTRICAL DATA (NOCT)

Maximum Power-P _{MAX} (Wp)	302	308	312	316	319	324
Maximum Power Voltage-V _{MPP} (V)	38.6	39.0	39.3	39.7	40.0	40.4
Maximum Power Current-I _{MPP} (A)	7.88	7.91	7.93	7.96	7.98	8.01
Open Circuit Voltage-V _{OC} (V)	46.6	47.0	47.5	47.8	48.2	48.7
Short Circuit Current-I _{SC} (A)	8.30	8.32	8.36	8.38	8.41	8.44

NOCT: Irradiance at 800W/m², Ambient Temperature 20°C, Wind Speed 1m/s.

MECHANICAL DATA

Solar Cells	Topcon Bifacial
No. of cells	144cells
Module Dimensions	1762x1134x30 mm (69.37x44.65x1.18 inches)
Weight	21.8kg (48.06 lb)
Front Glass	3.2 mm (0.12inches), High Transmission, AR Coated Heat Strengthened Glass
Encapsulant material	EVA/POE
BackSheet	Black Grid Transparent Backsheet
Frame	30mm (1.18 inches) Anodized Aluminium Alloy, Black
J-Box	IP 68 rated
Cables	Photovoltaic Technology Cable 4.0mm ² (0.006 inches ²), Landscape: N 1100 mm /P 1100 mm (43.31/43.31 inches)
Connector	MC4 Evo2
Fire Type	Type 1 or Type 2

TEMPERATURE RATINGS

NOCT (Nominal Operating Cell Temperature)	43°C (±2°C)
Temperature Coefficient of P _{MAX}	-0.30%/°C
Temperature Coefficient of V _{OC}	-0.25%/°C
Temperature Coefficient of I _{SC}	0.04%/°C

MAXIMUM RATINGS

Operational Temperature	-40~+85°C
Maximum System Voltage	1500V DC (IEC)
Max Series Fuse Rating	20A

WARRANTY

25 year Product Workmanship Warranty
25 year Power Warranty
1% first year degradation
0.4% Annual Power Attenuation

(Please refer to product warranty for details)

PACKAGING CONFIGURATION

Modules per box: 36 pieces
Modules per 40' container: 792 pieces



CAUTION: READ SAFETY AND INSTALLATION INSTRUCTIONS BEFORE USING THE PRODUCT.

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Version number: TSM_NA_2023_BV3

www.trinasolar.com



IQ8M and IQ8A Microinverters

Our newest IQ8 Microinverters are the industry’s first microgrid-forming, software defined microinverters with split-phase power conversion capability to convert DC power to AC power efficiently. The brain of the semiconductor-based microinverter is our proprietary application specific integrated circuit (ASIC) which enables the microinverter to operate in grid-tied or off-grid modes. This chip is built in advanced 55nm technology with high speed digital logic and has superfast response times to changing loads and grid events, alleviating constraints on battery sizing for home energy systems.



Part of the Enphase Energy System, IQ8 Series Microinverters integrate with the IQ Battery, IQ Gateway, and the Enphase App monitoring and analysis software.



Connect PV modules quickly and easily to IQ8 Series Microinverters using the included Q-DCC-2 adapter cable with plug-n-play MC4 connectors.



IQ8 Series Microinverters redefine reliability standards with more than one million cumulative hours of power-on testing, enabling an industry-leading limited warranty of up to 25 years.



IQ8 Series Microinverters are UL listed as PV Rapid Shutdown Equipment and conform with various regulations, when installed according to manufacturer’s instructions.

*Only when installed with IQ System Controller 2, meets UL 1741.
**IQ8M and IQ8A support split-phase, 240V installations only.

Easy to install

- Lightweight and compact with plug-n-play connectors
- Power Line Communication (PLC) between components
- Faster installation with simple two-wire cabling

High productivity and reliability

- Produce power even when the grid is down*
- More than one million cumulative hours of testing
- Class II double-insulated enclosure
- Optimized for the latest high-powered PV modules

Microgrid-forming

- Complies with the latest advanced grid support**
- Remote automatic updates for the latest grid requirements
- Configurable to support a wide range of grid profiles
- Meets CA Rule 21 (UL 1741-SA) and IEEE 1547:2018 (UL 1741-SB 3rd Ed.)

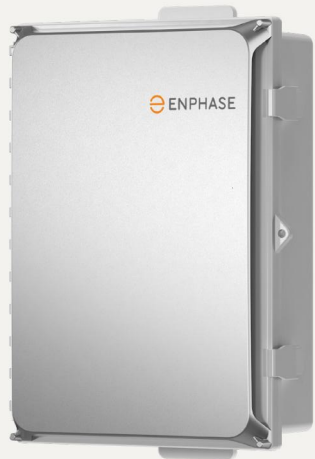
Note:

IQ8 Microinverters cannot be mixed together with previous generations of Enphase microinverters (IQ7 Series, IQ6 Series, etc) in the same system.

IQ8M and IQ8A Microinverters

INPUT DATA (DC)		I08M-72-2-US		I08A-72-2-US	
Commonly used module pairings¹	W	260 – 460		295 – 500	
Module compatibility		54-cell / 108 half-cell, 60-cell / 120 half-cell, 66-cell / 132 half-cell and 72-cell / 144 half-cell			
MPPT voltage range	V	30 – 45		32 – 45	
Operating range	V			16 – 58	
Min. / Max. start voltage	V			22 / 58	
Max. input DC voltage	V			60	
Max. continuous input DC current	A			12	
Max. input DC short-circuit current	A			25	
Max. module I _{sc}	A			20	
Overtoltage class DC port				II	
DC port backfeed current	mA			0	
PV array configuration		1 x 1 Ungrounded array; No additional DC side protection required; AC side protection requires max 20A per branch circuit			
OUTPUT DATA (AC)		I08M-72-2-US		I08A-72-2-US	
Peak output power	VA	330		366	
Max. continuous output power	VA	325		349	
Nominal (L-L) voltage / range²	V			240 / 211 – 264	
Max. continuous output current	A	1.35		1.45	
Nominal frequency	Hz			60	
Extended frequency range	Hz			47 – 68	
AC short circuit fault current over 3 cycles	Arms			2	
Max. units per 20 A (L-L) branch circuit³				11	
Total harmonic distortion				<5%	
Overtoltage class AC port				III	
AC port backfeed current	mA			30	
Power factor setting				1.0	
Grid-tied power factor (adjustable)				0.85 leading – 0.85 lagging	
Peak efficiency	%	97.8		97.7	
CEC weighted efficiency	%	97.5		97	
Night-time power consumption	mW			60	
MECHANICAL DATA					
Ambient temperature range		-40°C to +60°C (-40°F to +140°F)			
Relative humidity range		4% to 100% (condensing)			
DC Connector type		MC4			
Dimensions (H x W x D)		212 mm (8.3”) x 175 mm (6.9”) x 30.2 mm (1.2”)			
Weight		1.08 kg (2.38 lbs)			
Cooling		Natural convection – no fans			
Approved for wet locations		Yes			
Pollution degree		PD3			
Enclosure		Class II double-insulated, corrosion resistant polymeric enclosure			
Environ. category / UV exposure rating		NEMA Type 6 / outdoor			
COMPLIANCE					
Certifications	CA Rule 21 (UL 1741-SA), UL 62109-1, IEEE 1547:2018 (UL 1741-SB 3 rd Ed.), FCC Part 15 Class B, ICES-0003 Class B, CAN / CSA-C22.2 NO. 107.1-01 This product is UL Listed as PV Rapid Shutdown Equipment and conforms with NEC 2014, NEC 2017, and NEC 2020 section 690.12 and C22.1-2018 Rule 64-218 Rapid Shutdown of PV Systems, for AC and DC conductors, when installed according to manufacturer's instructions.				

(1) Pairing PV modules with wattage above the limit may result in additional clipping losses. See the compatibility calculator at <https://link.enphase.com/module-compatibility>. (2) Nominal voltage range can be extended beyond nominal if required by the utility.
(3) Limits may vary. Refer to local requirements to define the number of microinverters per branch in your area.



X-IQ-AM1-240-5
X-IQ-AM1-240-5C

IQ Combiner 5/5C

The IQ Combiner 5/5C consolidates interconnection equipment into a single enclosure and streamlines IQ Series Microinverters and IQ Gateway installation by providing a consistent, pre-wired solution for residential applications. IQ Combiner 5/5C uses wired control communication and is compatible with IQ System Controller 3/3G and IQ Battery 5P.

The IQ Combiner 5/5C, along with IQ Series Microinverters, IQ System Controller 3/3G, and IQ Battery 5P provides you with a complete grid-agnostic Enphase Energy System.



IQ Series Microinverters
The high-powered smart grid-ready IQ Series Microinverters (IQ6, IQ7, and IQ8 Series) dramatically simplify the installation process



IQ System Controller 3/3G
Provides microgrid interconnection device (MID) functionality by automatically detecting grid failures and seamlessly transitioning the home energy system from grid power to backup power



IQ Battery 5P
Fully integrated AC battery system. Includes six field-replaceable IQ8D-BAT Microinverters



IQ Load Controller
Helps prioritize essential appliances during a grid outage to optimize energy consumption and prolong battery life



5-year
limited
warranty



Smart

- Includes IQ Gateway for communication and control
- Includes Enphase Mobile Connect (CELLMODEM-M1-06-SP-05), only with IQ Combiner 5C
- Supports flexible networking: Wi-Fi, Ethernet, or cellular
- Provides production metering (revenue grade) and consumption monitoring

Easy to install

- Mounts to one stud with centered brackets
- Supports bottom, back, and side conduit entry
- Supports up to four 2-pole branch circuits for 240 VAC plug-in breakers (not included)
- 80 A total PV branch circuits

Reliable

- Durable NRTL-certified NEMA type 3R enclosure
- 5-year limited warranty
- Two years labor reimbursement program coverage included for both the IQ Combiner SKUs
- UL1741 listed

IQ Combiner 5/5C

MODEL NUMBER	
IQ Combiner 5 (X-IQ-AM1-240-5)	IQ Combiner 5 with IQ Gateway printed circuit board for integrated revenue grade PV production metering (ANSIC12.20 ±0.5%), consumption monitoring (± 2.5%) and IQ Battery monitoring (±2.5%). Includes a silver solar shield to deflect heat
IQ Combiner 5C (X-IQ-AM1-240-5C)	IQ Combiner 5C with IQ Gateway printed circuit board for integrated revenue grade PV production metering (ANSI C12.20 ±0.5%), consumption monitoring (±2.5%) and IQ Battery monitoring (±2.5%). Includes Enphase Mobile Connect cellular modem (CELLMODEM-M1-06-SP-05) ¹ . Includes a silver solar shield to deflect heat
WHAT'S IN THE BOX	
IQ Gateway printed circuit board	IQ Gateway is the platform for total energy management for comprehensive, remote maintenance and management of the Enphase IQ System
Busbar	125A busbar with support for 1 x IQ Gateway breaker and 4 x 20A breaker for installing IQ Series Microinverters and IQ Battery 5P
IQ Gateway breaker	Circuit breaker, 2-pole, 10 A/15 A
Production CT	Prewired revenue-grade solid core CT, accurate up to 0.5%
Consumption CT	Two consumption metering clamp CTs, shipped with the box, accurate up to 2.5%
IQ Battery CT	One battery metering clamp CT, shipped with the box, accurate up to 2.5%
CTRL board	Control board for wired communication with IQ System Controller 3/3G and the IQ Battery 5P
Enphase Mobile Connect (only with IQ Combiner 5C)	4G-based LTE-M1 cellular modem (CELLMODEM-M1-06-SP-05) with a 5-year T-Mobile data plan
Accessories kit	Spare control headers for CTRL board
ACCESSORIES AND REPLACEMENT PARTS (NOT INCLUDED, ORDER SEPARATELY)	
CELLMODEM-M1-06-SP-05	4G-based LTE-M1 cellular modem with a 5-year T-Mobile data plan
CELLMODEM-M1-06-AT-05	4G-based LTE-M1 cellular modem with a 5-year AT&T data plan
Circuit breakers (off-the-shelf)	Supports Eaton BR210, BR215, BR220, BR230, BR240, BR250, and BR260 circuit breakers Supports Eaton BR220B, BR230B, and BR240B circuit breakers compatible with hold-down kit
Circuit breakers (provided by Enphase)	BRK-10A-2-240V, BRK-15A-2-240V, BRK-20A-2P-240V, BRK-15A-2P-240V-B, and BRK-20A-2P-240V-B (More details in "Accessories" section)
XA-SOLARSHIELD-ES	Replacement solar shield for IQ Combiner 5/5C
XA-ENV2-PCBA-5	IQ Gateway replacement printed circuit board (PCB) for Combiner 5/5C
X-IQ-NA-HD-125A	Hold-down kit compatible with Eaton BR-B series circuit breakers (with screws)
ELECTRICAL SPECIFICATIONS	
Rating	80 A
System voltage	120/240 VAC, 60 Hz
Busbar rating	125 A
Fault current rating	10 kAIC
Maximum continuous current rating (input from PV/storage)	64 A
Branch circuits (solar and/or storage)	Up to four 2-pole Eaton BR series distributed generation (DG) breakers only (not included)
Maximum total branch circuit breaker rating (input)	80 A of distributed generation/95 A with IQ Gateway breaker included
IQ Gateway breaker	10 A or 15 A rating GE/Siemens/Eaton included
Production metering CT	200 A solid core pre-installed and wired to IQ Gateway
Consumption monitoring CT (CT-200-CLAMP)	A pair of 200 A clamp-style current transformers is included with the box
IQ Battery metering CT	200 A clamp-style current transformer for IQ Battery metering, included with the box

¹ A plug-and-play industrial-grade cell modem for systems up to 60 microinverters. (Available in the US, Canada, Mexico, Puerto Rico, and the US Virgin Islands, where there is adequate cellular service in the installation area.)

The Respect Your Roof Deserves

When integrating with a home, solar attachments must be dependable for the lifetime of the rooftop. Due to recent innovations, many asphalt shingles have bonded courses. A mount that protects without the need to pry shingles can really speed things up.

Halo UltraGrip™ (HUG®) is here to respect the roof. Its Halo is a cast-aluminum barrier that encases the UltraGrip, our industrial-grade, foam-and-mastic seal. This allows HUG to accelerate the installation process and provide the utmost in waterproofing protection. Give your roof a HUG®.



Rafter Mount



Deck Mount



Rafter & Deck Mounting Options

Mount HUG® to the roof rafters, the roof deck, or both with our custom-engineered RD (rafter-or-deck) Structural Screw. The RD Structural Screw anchors HUG® to the roof with an EPDM sealing washer, completing the stack of waterproofing barriers. See backside for more installation information.

Rated & Certified to Respect the Roof™
UL 2703, UL 2703A, 441 (27), TAS 100(A)-95

Adaptive, Rafter-Friendly Installation



Hit the rafter? Good to go!

When you find a rafter, you can move on. Only 2 RD Structural Screws are needed.



Miss the rafter? Try it again.

Place another screw to the left or right. If rafter is found, install 3rd and final screw.



Still no luck? Install the rest.

If more than 3 screws miss the rafter, secure six screws to deck mount it.

Trusted Strength & Less Hassle



Structural capacities of HUG® were reviewed in many load directions, with racking rail running cross-slope or up-slope in relation to roof pitch.

For further details, see the HUG® certification letters for attaching to rafters and decking.

IronRidge designed the HUG®, in combination with the RD Structural Screw to streamline installs, which means the following:

- No prying shingles
- No roof nail interference
- No pilot holes necessary
- No sealant (in most cases)
- No butyl shims needed

Attachment Loading



The rafter-mounted HUG® has been tested and rated to support 1004 (lbs) of uplift and 368 (lbs) of lateral load.

Structural Design



Parts are designed and certified for compliance with the International Building Code & ASCE/SEI-7.

Water Seal Ratings



HUG® passed both the UL 441 Section 27 "Rain Test" and TAS 100(A)-95 "Wind Driven Rain Test" by Intertek.

UL 2703 & 2703A



Systems conform to UL 2703 and HUG® also conforms to UL 2703A. See Flush Mount Manual for more info.



Aire® Flush Mount System



Breathe easy with accelerated installations.

The Aire® racking system has been carefully engineered to streamline every part of the installation process. We've eliminated tiresome hassles, so that you get off the roof and on to your next project faster than ever.

Aire® retains the strength and reliability that IronRidge installers depend on. It also takes wire management to the next level with the first (and only) NEC-compliant rail, formally approved and listed as a cable tray.



Strength Tested

All components have been evaluated for superior structural performance.



Class A Fire Rating

Certified to maintain the fire resistance rating of the existing roof structure.



UL 2703 Listed System

Entire system and components meet the latest effective UL 2703 standards.



PE Certified

Pre-stamped engineering letters are available online for most states.



Approved Cable Tray

Open channel listed to NEMA VE 1, certified to hold PV and DG cables.

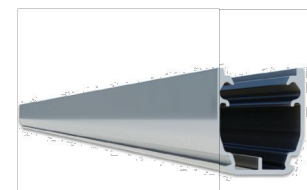


25-Year Warranty

Products are guaranteed to arrive without any impairing defects.

Rails

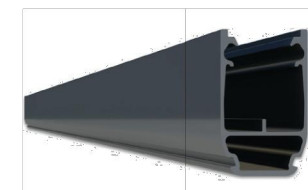
Aire® A1 Rail



The lighter, open Aire® rail for standard conditions.

- 6' spanning capability
- Wire management tray
- Mill or anodized black

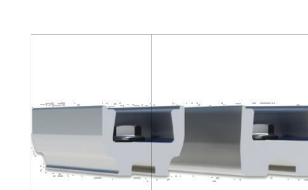
Aire® A2 Rail



The tougher, open Aire® rail for higher load capacity.

- 8' spanning capability
- Wire management tray
- Mill or anodized black

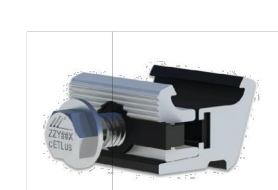
Aire® Rail Ties



Structurally connect and bond Aire® Rails together.

- Reinstallable, up to 5x
- Internal splice design
- No more splice rules

Aire® Dock



Connects Aire® Rails to attachments with ease.

- Clicks on, slides easily
- Drops into open slots
- Anodized assembly

Clamps & Grounding

Aire® Lock Mids



Securely bond between modules to Aire® Rails.

- Fits 30-40mm modules
- Utilizes UFO® design
- Minimal 1/2" gap

Aire® Lock Ends



Securely bond modules to Aire® Rails along ends.

- Fits 30-40mm modules
- Easy rail engagement
- Clean aesthetics

Aire® Lock Stealth®



Securely bonds modules to rail ends, entirely hidden.

- Angled for easy install
- Robust tether leash
- Fits most modules

Aire® Lug



Bonds Aire® Rails to grounding conductors.

- Simplified with single bolt
- Low-profile form factor
- Works with 10-6 AWG

Accessories

Aire® Caps



Block entry and provide a finished look to Aire® Rails.

- Stay secure on rail ends
- Symmetrical, with drain
- Cover rough-cut ends

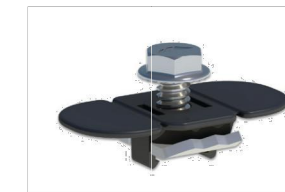
Aire® Clip



Keeps wiring contained in open Aire® Rail channels.

- No module interference
- Simple press-in design
- Slot for easy removal

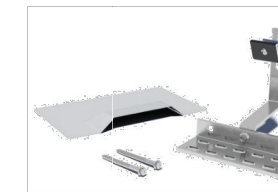
Aire® MLPE Mount



Securely bonds MLPE and accessories to Aire® Rails.

- Glove-friendly installation
- Lays flush in rail channel
- Low profile form factor

Aire® All Tile Hook



Attaches rails to tile roofs, with Aire® Dock included.

- Works on flat, S, & W tiles
- Single-socket installation
- Optional deck flashing

Resources



Design Assistant

Quickly go from rough layout to fully engineered system.

Go to [IronRidge.com/design](https://www.ironridge.com/design)



Approved for FL Hurricane Zones

Aire® has Florida Product Approval. Additional details can be found on the Florida Building Code website.

Learn More at bit.ly/florida-aire



28357 Industrial Blvd.
Hayward, CA 94545
1-800-227-9523
IronRidge.com

Attn: Sean McDonald, CEO, IronRidge Inc.
Date: February 27th, 2025

Re: Engineering Certification for the IronRidge Design Assistant – XR and AIRE Pitched Roof Systems

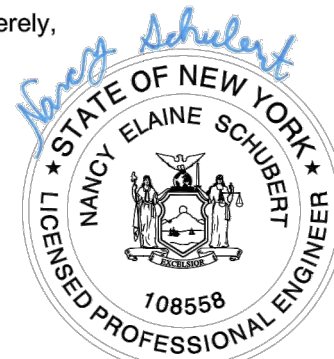
The IronRidge Design Assistant (DA) software includes structural analysis for the IronRidge XR system (XR10, XR100 and XR1000 rails) and AIRE system (A1 and A2 Rails), when used in pitched roof applications. The IronRidge XR and AIRE Flush Mount Systems are proprietary rooftop mounting systems used to support photovoltaic (PV) modules installed in portrait or landscape orientation and set parallel to the underlying roof surface. PV modules are supported by extruded aluminum rails and secured to the rails with IronRidge mounting clamps. The rails are side mounted to a selected roof attachment with stainless steel bonding hardware and the attachment is secured directly to the underlying roof structure.

This letter certifies that the IronRidge Design Assistant, used for the structural analysis of the IronRidge XR or Aire Flush Mount Systems, complies with the following codes and design criteria. When the IronRidge XR and AIRE flush mount systems are installed according to the IronRidge DA project report and manufacturer's installation guidance, compliance with below codes and design criteria is met.

Building Codes:

1. 2020 Building Code of New York State
2. 2020 Residential Code of New York State
3. ASCE/SEI 7-10, 7-16, 7-22, Minimum Design Loads for Buildings and Other Structures, by American Society of Civil Engineers
4. International Building Code, 2012-2024 Edition
5. International Residential Code, 2012-2024 Edition
6. AC428, Acceptance Criteria for Modular Framing Systems Used to Support Photovoltaic (PV) Panels, November 1, 2012 by ICC-ES
7. Aluminum Design Manual 2015 & 2020, by The Aluminum Association, Inc.
8. ANSI/AWC NDS-2015 & 2018, National Design Specification for Wood Construction, by the American Wood Council
9. SEAOC (Structural Engineer Association of California) report PV2-2017 Wind Design for Solar Arrays

Sincerely,



Expires: 9/30/2026

Nancy Elaine Schubert, P.E.

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It is a violation of §7209.2.a of Article 145 of the State Education Law for any person to alter any document that bears the seal of a Professional Engineer, unless the person is acting under the direction of a licensed Professional Engineer.